Experiment - 1

Manual Calculations:-

A. 8 bit Addition & Subtraction with

Flag Analysis:

* 9CH + 7AH

= 1001 1100 + 0111 1010

= 1 0001 0110 → A = 16H, CY = 1

Lower Nibble: C + A = 16H → AC = 1.

16H has two 1s → P = 1.

Signed: -100 + 122 = 22 → no overf

Flag bytes = 0001 0001 = 11H.

* 9CH - 7AH

= 22H, CY = 0.

22H has two 1s → P = 1.

Lower nibble: CH - AH = 2H → AC

Signed: -100 - (+122) = -222,

outside -128 to +127 → overf

Flag bytes = 0001 0100 = 14H

B. 16bit Addition & Subtraction.

* 1234H + 5678H

= 68ACH → no carry, CY = 0

```
 1234
+5678
------
 68AC
======
```

Lower byte:

34H + 78H

```
 0011  0100
+0111  1000
-----------
1010  1100 = ACH
```

Carry out = 0

Upper byte:

12H + 56H + 0

```
 0001  0010
+0101  0110
-----------
0110  1000 = 68H
```

Carry out = 0

∴ Result = 68AC(H), CY = 0

PDF Image to PDF

C. 8-bit Division.

* 64H ÷ 05H

Convert to binary:

64H = 0110 0100

05H = 0000 0101

Repeated subtraction.

```
  0110  0100
- 0000  0101
-------------
  0101  1111
```

```
  0101  1111
- 0000  0101
-------------
  0101  1010
```

~~0101 1010~~

```
  0101  1010
- 0000  0101
-------------
  0101  0101
```

```
  0101  0101
- 0000  0101
-------------
  0101  0000
```

The subtraction continues until

division cannot be subtracted further.
After 20 subtractions.

0000 0000

Therefore:

Quotient = 20 decimals

= 0001 0100

= 14H.

Remainder = 0000 0000

= 00H

Final:

Q = 14H , R = 00H.

D. 16-bit Division:

5678H ÷ 1234H.

Convert to Binary:

5678H = 0101 0110 0111 1000

1234H = 0001 0010 0011 0100

1st subtraction:

$$\begin{array}{r} 0101\ 0110\ 0111\ 1000 \\ -\ 0001\ 0010\ 0011\ 0100 \\ \hline 0100\ 0100\ 0100\ 0100 \end{array}$$

2nd Subtraction:-

```
  0100   0100   0100   0100
- 0001   0010   0011   0100
--------------------------
  0011   0010   0001   0000
```

3rd subtraction:

```
  0011   0010   0001   0000
- 0001   0010   0011   0100
--------------------------
  0001   1111   1101   1100
```

4th subtraction:

```
  0001   1111   1101   1100
- 0001   0010   0011   0100
--------------------------
  0000   1101   1010   1000
```

Now:

```
  0000   1101   1010   1000
<
  0001   0010   0011   0100
```

so another subtraction is not possible.

Therefor:

Quotient = 4

= 0000 0000 0000 01000

= 0004H

Remainder = 0000 1101 1010 1000

= 0DA8H

Final:

Q = 0004H , R=0DA8H.

Verification

4 × 1234H + 0DA8H = 5678H.

PDF Image to PDF